

Cross-Validation for Patient-Level Data

Why Leave-One-Subject-Out matters for EEG, and what it actually protects against

1. The problem ordinary cross-validation doesn't catch

In typical machine learning, k-fold cross-validation randomly splits all your data into groups, trains on some, and tests on the rest, repeating a few times. This works well when each data point is independent of the others. EEG epochs are not independent in that way: every person's brain activity has its own baseline rhythm, noise pattern, and idiosyncrasies. If windows from the same patient appear in both the training set and the test set — even from different moments in time — a model can quietly learn to recognize *the patient*, rather than the thing you actually want it to detect (a seizure).

This failure mode is dangerous specifically because it's invisible in the evaluation numbers: the model can score very well on this kind of split while having learned almost nothing that would generalize to a genuinely new patient.

2. What Leave-One-Subject-Out actually does

Leave-One-Subject-Out (LOSO) cross-validation fixes this by holding out an entire patient's data at a time — every epoch from that person is set aside as the test set, and the model trains on everyone else. This is repeated once per patient, so each person is tested on exactly once, always as someone the model has never seen during training for that fold.

With 24 patients, this produces 24 separate training-and-testing rounds (24 'folds'), each one asking the same question: *if this model had never seen this specific person before, how well would it actually do on them?* That is a far more honest test of real-world generalization than any random split could offer.

3. How this is implemented (conceptually, not in code)

Every feature vector needs to be tagged with which patient it came from — commonly called the 'groups' array. A cross-validation tool built for this purpose (such as scikit-learn's `LeaveOneGroupOut`) uses that tagging to automatically construct each fold: it looks at how many distinct patients exist in the groups array, and produces exactly that many train/test splits, each one holding out a different patient entirely.

A useful debugging instinct worth internalizing: if you ever see fewer folds run than the number of unique patients in your data, something is short-circuiting the loop early — commonly a subtle indentation mistake, where code meant to run *after* a loop finishes is accidentally placed *inside* it, causing the function to exit after only the first iteration.

4. Reading the results properly

Accuracy alone can be misleading, especially with imbalanced data (many more non-seizure epochs than seizure ones in a real dataset). A model that always predicts 'no seizure' can still score high accuracy while being clinically useless. This is why precision, recall, and F1-score matter alongside accuracy:

- **Precision:** of everything the model flagged as a seizure, how much actually was one? Low precision means too many false alarms.
- **Recall:** of all the real seizures in the test set, how many did the model actually catch? Low recall means missed seizures — usually the more clinically costly mistake.
- **F1-score:** a single number balancing precision and recall together, useful for comparing models at a glance without picking one metric to prioritize arbitrarily.

Why this generalizes directly to your PhD work

This is not a side detail specific to a classical baseline — cross-subject generalization is arguably the central open challenge in EEG-based seizure and SOZ localization research generally. A model that only works on patients it was trained on has limited clinical or scientific value. Every module still ahead in this roadmap — the CNN, the autoencoder, the diffusion-based augmentation, and the final three-stage architecture — should ultimately be evaluated the same way: by asking whether it holds up on a patient it has never seen.